To Share Textual Structure Globally: Development of TEI Viewer for East Asian Texts

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Introduction

The Text Encoding Initiative (TEI) Guidelines, an XML-based standard established in 1987 by European and North American humanities scholars, were developed to structure textual materials so they can be more effectively shared and utilized as research data. Over time, the TEI Guidelines have been continuously updated to accommodate evolving scholarly needs, influenced by both technological advancements and increasing internationalization. Since 2016, the focus on broader geographic regions has grown, with specialized groups dedicated to East Asia, Japan, and India. These efforts led to the incorporation of rules for Japanese ruby annotations, reflecting a commitment to address region-specific textual features.

Despite these developments, practical difficulties remain in certain contexts, such as Japan, where TEI/XML-compliant data has not yet been fully integrated into scholarly workflows. While tutorials in Japanese have been created to guide the encoding of TEI/XML data, the subsequent steps—effectively using and leveraging that data—pose significant challenges. This is largely due to the specialized programming skills often required, especially knowledge of XSLT, which is not widely taught or documented in Japanese.

In European and North American digital humanities curricula, TEI/XML is frequently taught alongside XSLT, allowing scholars to confidently extract, transform, and analyze their data. However, in Japan, where XSLT and relevant programming skills are less common, scholars struggle to move beyond basic encoding. Many humanities researchers are interested in the encoding process to preserve their interpretations but find programming tasks discouraging or time-consuming. This lack of accessible tools hampers motivation and skill development.

Furthermore, existing TEI-compliant viewers such as CETEICean, TEI Boilerplate, Versioning Machine, and TEI Critical Apparatus Toolbox do not support vertical writing systems or the complex page layouts typical of East Asian classical texts. These limitations prompted the creation of a specialized TEI/XML-compatible viewer, designed to display encoded texts more intuitively and align with the unique needs and scholarly practices found in East Asian textual studies.

Development Policy

In the prototype phase, we tried two approaches: loading TEI/XML files directly into a web browser with JavaScript, and using Python to convert them into HTML+JavaScript. Initially based on CETEIcean, we eventually shifted to a custom codebase.

Our aim was to allow those performing TEI/XML markup to easily confirm the appearance of their files. CETEIcean's requirement for HTTP connections posed a setup hurdle for many Japanese humanities scholars. Thus, we used the File API to load local TEI/XML files directly into the browser, eliminating the need for servers.

We first focused on classical Chinese Buddhist texts widely read across East Asia, particularly their critical apparatus. Later, to broaden applicability, we collaborated with researchers from fields including Heian literature, waka poetry, manuscript studies, modern letters, modern novels, ancient history, and premodern Buddhist exegesis. These collaborations guided us in developing features aligned with real-world needs.

In January 2024, we released an integrated version that lets users load TEI/XML files directly into the browser (Fig.1). Initially employing jQuery and the File API, we ensured offline usability by avoiding external libraries, except OpenStreetMap. Distributed under an open license, it can be downloaded and freely used. Our website presents TEI/XML files produced in joint research as examples. By September 9, 2024, key features include:

- Display of basic TEI structures and vertical text
- Critical apparatus display (parallel-segmentation and double-end-point attached)
- Text reconstruction from collation data
- IIIF-compatible image display via <zone>
- Highlighting of ID-referenced relationships
- Graph visualization of references (D3.js)
- Listing and displaying proper nouns
- Configurable addition of ID-referenced elements
- Switching <choice> elements (sic|corr, orig|reg)
- Toggle <front><body><back> sections
- Adjustable text area height by dragging
- Displaying <place> with OpenStreetMap

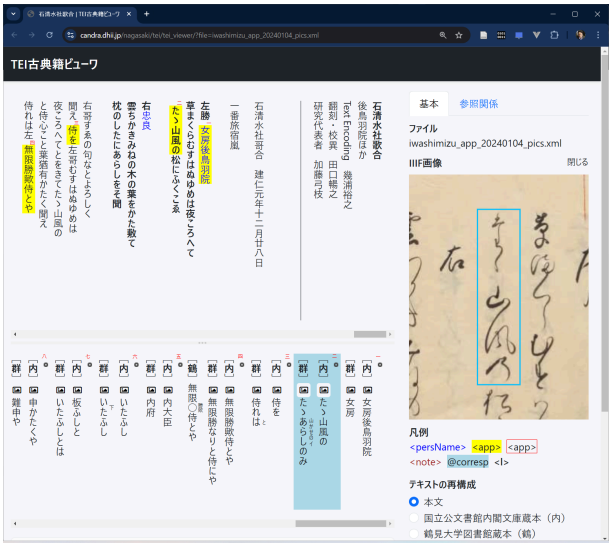

Fig.1 An example of showing a variant readding with a part of a IIIF-compliant image

Outcomes

This integrated TEI classical text viewer for East Asian texts has simplified the structured display of East Asian texts. Alongside a Japanese TEI primer published in 2022, it has fostered adoption in university courses and seminars, expanding the user community. In November 9, 2024, TEI Guidelines was seriously discussed at the Waka Literature Society, a traditional humanities association.

The open license has enabled the development of more advanced viewers, expanding possibilities for humanities text data use. Although the necessity of programming remains debatable in digital humanities, this approach—focusing on sharing and preserving humanities research—is highly beneficial. It can serve as an effective model in regions where TEI is still unfamiliar, promoting the sharing of textual structures as research data.

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